AI251P Applied Computer Vision

Capstone Project – Phase 1 Report

Smart Pet Vision: Cat and Dog Classification, Detection and Segmentation for Adoption-Listing Photographs

Smart Pet Vision · Phase 1 (M1–M3, 40 points) · Due 28 October 2026

1 Team and project

Field	
Team name	Smart Pet Vision
Topic	Original topic, approved by the instructor (approval confirmed by 19 October 2026)
Application area	Animal welfare: preparing cat and dog photographs for adoption listings
Code repository	[https://github.com/Yasser-20-alhajlah/smart-pet-vision] (https://github.com/Yasser-20-alhajlah/smart-pet-vision) faf05893291e9c1209cd07d63d891bc0b8acc2de
W&B project	vandb. ai/yasser-alhaj lah- kaust/smart- pet-vision; summary report: W&B report
Dataset card	Appendix A (the same text is docs/dataset_card. md in the smart-pet-vision folder)

Team members

Name	Student ID	Workstream (Table 3)	Phase 1 contribution
Abdulrahman Mastour Alharthi	225478	WS1 Data and classification	Data cleaning, split and audits (M1); the three classifiers and their evaluation (M2–M3); final edit of this report and the evidence folders
Yasser Mansour Alhajlah	225481	WS2 Detection	Body boxes derived from the trimaps and the box review; W&B project; full reproduction on a clean runtime (23 October)
Mohammed Adeeb Alrashdi	225488	WS3 Segmentation	Masks derived from the trimaps; annotation guide for the team photos
Musaad Obaied Saad Alqahtani	225471	WS4 Integration	Repository, single configuration and W&B set-up; reproducibility instructions

All four members reviewed and corrected about 100 team photos each in CVAT.

Problem and acceptance criteria

- **Summary:** The system takes one adoption-listing photograph and decides whether it is ready to publish or should be referred to shelter staff. It reports the species with a temperature-scaled confidence score, the body box and mask, and a cut-out of the animal; Phase 1 builds the data foundations and the species classifier.
- **Input → output:** one photograph of a cat or a dog → species and confidence, body box, mask and its area share, cut-out, and a verdict (ready for the listing, or refer to staff).

- **Intended users and the decision supported:** staff of animal shelters and adoption platforms; they decide whether a photo is published automatically or checked by a person. A photo is referred when the classifier's confidence is below the threshold, when the classifier and the detector disagree on the species, or when no animal is found. The confidence threshold is provisional until it is measured on an independent calibration set (Section 3.4).
- **Acceptance criteria (fixed before results):** Phase 1: test accuracy ≥ 0.90 and test macro-F1 ≥ 0.90 for the selected classifier. Phase 2 (for reference): mAP50 ≥ 0.75, recall ≥ 0.80, mIoU ≥ 0.75, Dice ≥ 0.85, service success rate ≥ 99%, p95 latency ≤ 2 s on a T4, ONNX parity ≤ 0.001.

2 Data foundations (M1) (12 points)

2.1 Datasets

Dataset	Link / ref.	Content and annotations	Use	Licence	Images
Oxford-IIIT Pet	VGG, Oxford; Parkhi et al. (2012)	37 breeds (12 cat, 25 dog); species and breed labels, pixel trimaps, head boxes (body boxes derived from the trimaps)	Train / val / test	CC BY-SA 4.0	7,349
Team-collected (bonus)	Folder `SmartPetVision_TeamData` in the submission archive, with a README (Section 2.6)	Photos of cats and dogs supplied by team members (104 of them taken by the pets' owners); labels, boxes and polygon masks annotated by the team in CVAT (503 animals)	External test only	Consent: yes	401

The full Oxford-IIIT Pet set is used; no subset was selected. Of 414 team photos received, 401 were kept (Section 2.6).

2.2 Cleaning

Check	Method / tool	Images removed
Corrupt or unreadable files	Pillow open and verify; minimum side 32 px	0
Exact duplicates	MD5 of the file bytes	0
Near-duplicates	pHash, Hamming distance ≤ 4: 45 pairs merged by union-find into 41 groups (84 images, largest group 3); groups are kept whole in one split rather than removed	0
Other (anonymisation, out-of-scope images, . . .)	Empty trimap (no usable mask or body box)	14
Images remaining	2,366 cats, 4,969 dogs, 37 breeds; median size 500 × 375 px	**7,335**

2.3 Leakage-safe split

- **Grouping key:** near-duplicate group (pHash Hamming distance ≤ 4, merged transitively), so every image and its near-copies form one group; 7,292 groups in total.
- **Why this prevents leakage:** Oxford-IIIT Pet contains near-identical shots of the same pet (burst photos, crops and re-encodings). Splitting image by image would put such copies in train and test and inflate test scores. Audit after the split: groups crossing splits = 0; MD5 matches crossing splits = 0. The dataset has no identifier for the individual animal, so different poses of the same pet can still cross splits. This identity leakage was *measured* (table below): its effect on the selected model's test score is at most 0.001.
- **Stratification and seed:** 70/15/15 by image count, stratified by breed at group level (`data.py`, `group_stratified_split`); seed 42.

Split	No. of Groups	No. of Images	Images per class (or % of the smallest class)
Train	5,091	5,123	cat 1,652 (32.2%), dog 3,471; smallest breed 128 images
Validation	1,103	1,106	cat 357 (32.3%), dog 749; smallest breed 28 images
Test	1,098	1,106	cat 357 (32.3%), dog 749; smallest breed 28 images

Identity-leakage measurement (notebook `02b_phase1_supplement`; files `runs/supplement/identity_*`). Without identity labels, every test image is paired with its most similar *training* image of the same breed (cosine similarity of ImageNet ResNet-50 features, a network not trained on this data; median 0.80, 99th percentile 0.955, maximum 0.98). If the same animal in train and test inflated the scores, accuracy would fall when the test images with a close training match are removed. It does not for the selected model:

Test images kept	Removed (close match)	n	Fine-tuned accuracy (95% CI)	Scratch accuracy (95% CI)
All	0	1,106	0.999 (0.995–1.000)	0.906 (0.887–0.922)
Similarity < 0.95	22 (2.0%)	1,084	0.999 (0.995–1.000)	0.905 (0.886–0.921)
Similarity < 0.90	133 (12.0%)	973	0.999 (0.994–1.000)	0.902 (0.882–0.919)
Similarity < 0.85	308 (27.8%)	798	0.999 (0.993–1.000)	0.892 (0.869–0.912)
Similarity < 0.80	562 (50.8%)	544	0.998 (0.990–1.000)	0.884 (0.855–0.908)
Similarity < 0.75	790 (71.4%)	316	0.997 (0.982–0.999)	0.851 (0.808–0.886)

Test split; input 224 × 224; argmax decision, no threshold; Wilson intervals. The fine-tuned model makes one error in every subset. Removing the 22 closest pairs (the most likely same-animal cases) leaves both models unchanged. The scratch CNN drops gradually only when most of the test set is removed; the remaining images are the least typical of their breed, so this is read as difficulty, not leakage. Similarity is a proxy for identity, so identity leakage is bounded, not excluded; the 40 most similar pairs are rendered in `identity_top_pairs.jpg` for visual review.

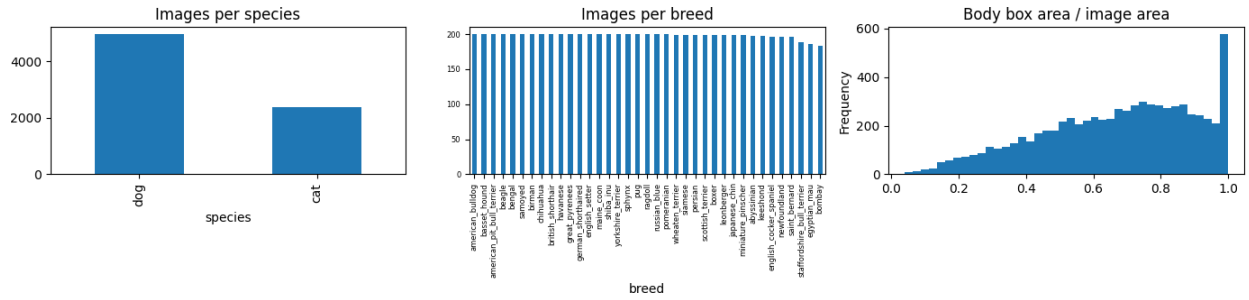


Figure 1: Images per species, images per breed, and body-box area as a fraction of image area (all 7,335 cleaned images; original resolution; boxes derived from the trimaps; no threshold).

2.4 Label audit

Audited sample size	Errors found	Estimated label-error rate	Upper bound (95% confidence)
200 images (species label)	0	0.0%	1.5% (one-sided, rule of three)
Same 200 (breed label)	4 doubtful	2.0%	5.0% (Wilson interval 0.8–5.0%)
300 images (derived body box)	4 loose boxes	1.3%	3.4% (Wilson interval 0.5–3.4%)

Both samples were drawn at random with the project seed and rendered as contact sheets (`data. py`, `reviev_sheets`). The review was done visually with AI assistance and confirmed by the team; the reviewer and confirmation are recorded per image in `runs/data/reviev/labels/reviev. csv` and `runs/data/reviev/boxes/reviev. csv`, and notebook 01 prints the summary from these files. No species error was found. The four doubtful breed labels look like another breed or a mix (for example, a 'beagle' that looks like a Jack Russell terrier). The four flagged boxes are all loose: the trimap includes background around a small animal, and none of them misses the animal.

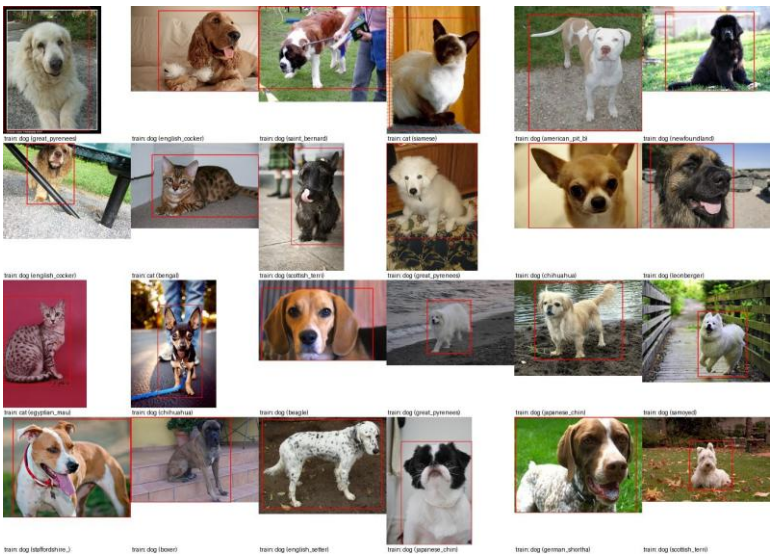


Figure 2: Visual check of the body boxes derived from the trimaps (training-split samples; original resolution; no threshold).

2.5 Data loader and augmentation

Input resolution 224 × 224 px; ImageNet mean and standard deviation for normalisation; batch 64; the training loader is shuffled with a seeded generator (`torch.Generator`, seed 42).

- **RandomResizedCrop** to 224 px, scale 0.5–1.0: adoption photos are framed loosely and, in Phase 2, the classifier receives detector crops; random crops simulate both.
- **Random horizontal flip** ($p = 0.5$): a mirrored cat or dog is still the same species, and pets face both ways.
- **ColorJitter** (brightness, contrast and saturation 0.2; hue 0.02): shelter photos vary in lighting and camera; the hue change is kept small so that coat colour, which is informative, is preserved.
- **Validation and test:** resize to 255 px and centre crop 224 px, without augmentation.
- **Class imbalance** (about two dogs per cat) is handled by breed stratification and by macro-F1 as the primary metric; no class weights or resampling.

2.6 Dataset card summary

- **Class imbalance:** about 2:1 dogs to cats; breeds are roughly balanced (smallest test breed 28 images).
- **Domain bias:** web and pet-site photos, mostly one large, centred, well-lit animal (1,105 of 1,106 test boxes are 'large'). Small, partly visible and multiple animals are under-represented, which is where shelter photos differ. ImageNet and Oxford-IIIT Pet were both collected from the web, so some overlap between pretraining images and the test split cannot be excluded.
- **Annotation quality:** species labels audited (0 of 200 errors, upper bound 1.5%); body boxes are derived from trimaps, not drawn, and 1.3% are loose; the same animal may appear in more than one split (no identity labels); its measured effect on the selected model is at most 0.001 (Section 2.3).
- **Licence:** Oxford-IIIT Pet is CC BY-SA 4.0; images are downloaded from the official source and not redistributed.
- **Privacy and storage (team photos):** kept outside the code repository in the folder `SmartPetVision_TeamData`, with its own README (included in the submission archive for the instructor only and not redistributed); unprocessed originals, which may carry camera metadata, are kept privately in a separate folder and never shared. Owner consent for all 401 kept photos; EXIF orientation applied and all metadata, including GPS, removed; 13 photos excluded as suspected stock or generated images; MD5 and pHash checked within the set and against Oxford-IIIT Pet (no duplicates). Annotation: an automatic draft was reviewed and corrected in CVAT by the four team members, about 100 photos each; eight species labels were corrected. The set is never used for training, thresholds or model selection.

3 Classification (M2–M3) (20 points)

Task: two-class species classification (cat, dog) of whole images from Oxford-IIIT Pet. In Phase 2 the selected model classifies the detector's crop; its confidence drives the low-confidence referral, and its species is cross-checked against the detector's class (a disagreement is referred to staff).

3.1 Runs compared

Models are selected on the validation split only, by validation macro-F1 at the best epoch; the test split is evaluated once, for the final models. Common recipe: AdamW (weight decay 10^{-4}), cosine schedule, batch 64, mixed precision, cross-entropy, seed 42; temperature scaling fitted on the validation split. The scratch CNN has five blocks of two 3 × 3 convolutions with BatchNorm and ReLU, each followed by max-pooling (32 to 256 channels), global average pooling, dropout 0.3 and a linear layer (2,354,402 parameters). The frozen model trains only a new 2048 → 2 head (4,098 parameters) with BatchNorm statistics frozen; full fine-tuning trains all 23.5 M parameters from the ImageNet weights with a ten times smaller learning rate. All runs are in W&B group `full-rerun-2026-10-18`.

Run (W&B link)	Backbone	Training	Input res.	Epochs	Val. acc.	Val. macro-F1
`gh7ayx6h`	Custom CNN (10 conv.)	From scratch (lr = 10^{-3})	224	25 (best 25)	0.928	0.918
`xsz5afr9`	ResNet-50	Pretrained, frozen; linear probe (10^{-3})	224	8 (best 6)	0.998	0.998
`8nvxbsn2`	ResNet-50	Pretrained, full fine-tuning (10^{-4})	224	10 (best 6)	1.000	1.000

Test-set results (final models only)

Run	Test accuracy (95% CI)	Test macro-F1	ECE (raw → calibrated)
From scratch	0.906 (0.887–0.922)	0.892	0.014 → 0.025
Pretrained, frozen	0.999 (0.995–1.000)	0.999	0.011 → 0.001
Fine-tuned (selected)	0.999 (0.995–1.000)	0.999	0.0015 → 0.0009

Test split (n = 1,106), input 224 × 224 px, decision = argmax of the two class probabilities (no threshold). Intervals are Wilson 95%. ECE uses 15 bins.

Which model you keep and why: the fully fine-tuned ResNet-50, chosen by the rule fixed in advance (best validation macro-F1: 1.000 against 0.998 for the frozen model, about two validation images). It meets both Phase 1 criteria (0.999 / 0.999); the scratch CNN meets the accuracy criterion but not macro-F1 (0.892). The two transfer models are statistically indistinguishable on the test split (exact McNemar test: one discordant error each, $p = 1.0$), while both differ from the scratch CNN ($p < 10^{-28}$); the frozen model is an equally valid choice at a fraction of the training cost.

3.2 Learning curves

Same training, fully logged. The submitted runs logged training loss and validation scores only, so the three trainings were repeated with the same configuration, seed and order, now also logging training accuracy and validation loss (the added metrics draw no random numbers). The rerun reproduced every previously logged value exactly (training loss, validation accuracy and macro-F1 at every epoch of all three models; `runs/supplement/curves/rerun_vs_submitted.csv`), so these curves belong to the submitted training.

Scratch CNN: under-trained, not over-fitted. At epoch 25 training accuracy (0.914) is still below validation accuracy (0.928) and validation loss (0.193) is below training loss (0.204) and still falling; both gaps are consistent with augmentation and dropout acting on training batches only. There is no sign of over-fitting; more epochs or a learning-rate warm-up are the first things to try. The validation spikes at epochs 6 and 9 (validation loss 0.89 and 1.70 while training loss keeps falling) show unstable evaluation-mode BatchNorm statistics at the high initial learning rate. **Fine-tuned**

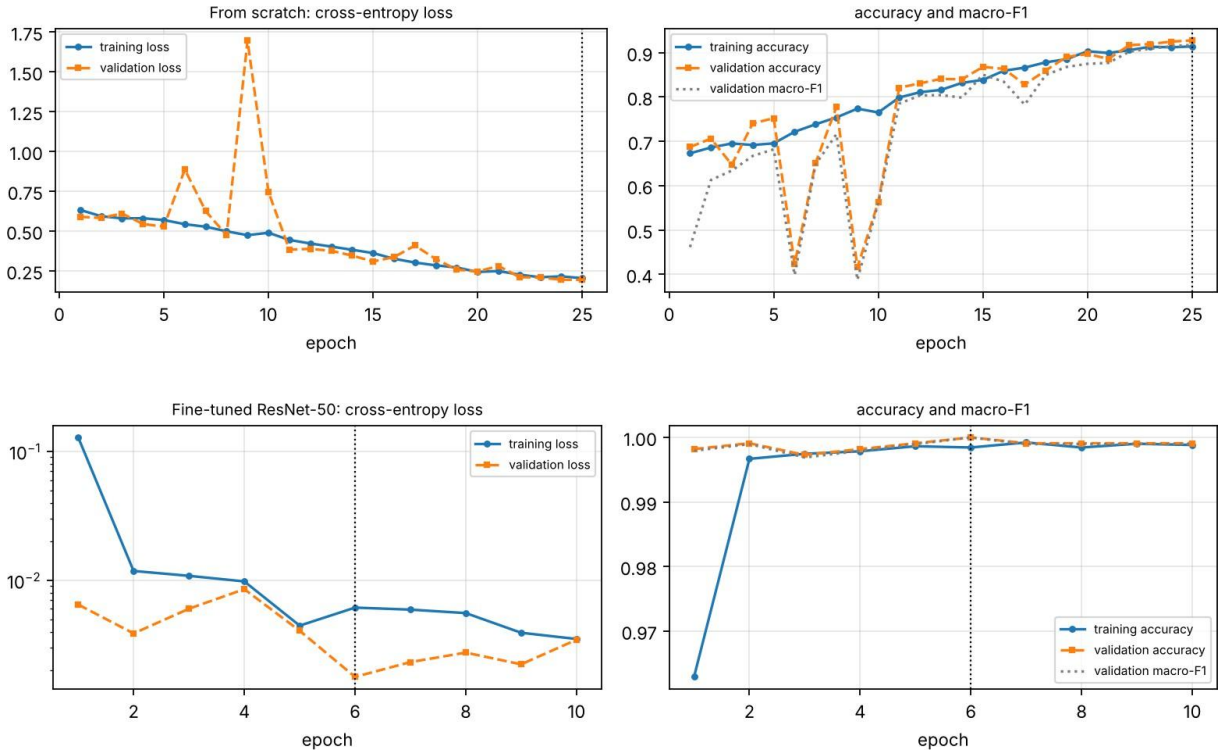


Figure 3: Training and validation loss (left) and training and validation accuracy with validation macro-F1 (right) per epoch: (top) from scratch, (bottom) fine-tuned (log loss axis). Validation split (n = 1,106); input 224 × 224 px; argmax decision, no threshold. Training metrics are running averages over augmented batches in train mode (dropout on). The dotted line marks the kept checkpoint. From the full-logging rerun (notebook `02b`; W&B `bvns5vir` scratch, `4ixtdkiq` frozen, `b3t0emrs` fine-tuned).

ResNet-50: validation loss reaches its minimum (0.0018) at epoch 6, the kept checkpoint, then rises to 0.0035 by epoch 10 while validation accuracy stays at 0.999: mild over-fitting after epoch 6, which the checkpoint rule avoids (best validation macro-F1 and minimum validation loss fall on the same epoch). **Frozen model:** validation loss falls at every epoch (0.063 → 0.013); no over-fitting.

3.3 Per-class results (selected model, test set)

Class	Precision	Recall	F1	Support
Cat	1.000	0.997	0.999	357
Dog	0.999	1.000	0.999	749
Macro average	0.999	0.999	0.999	1,106

Top-k accuracy ($k = 2$): 1.000, which is trivial with two classes; top-1 accuracy (0.999) is the informative figure. For comparison, the scratch CNN reaches cat 0.856 / 0.852 / 0.854 and dog 0.929 / 0.932 / 0.931 (precision / recall / F1).

3.4 Confusion matrix and calibration

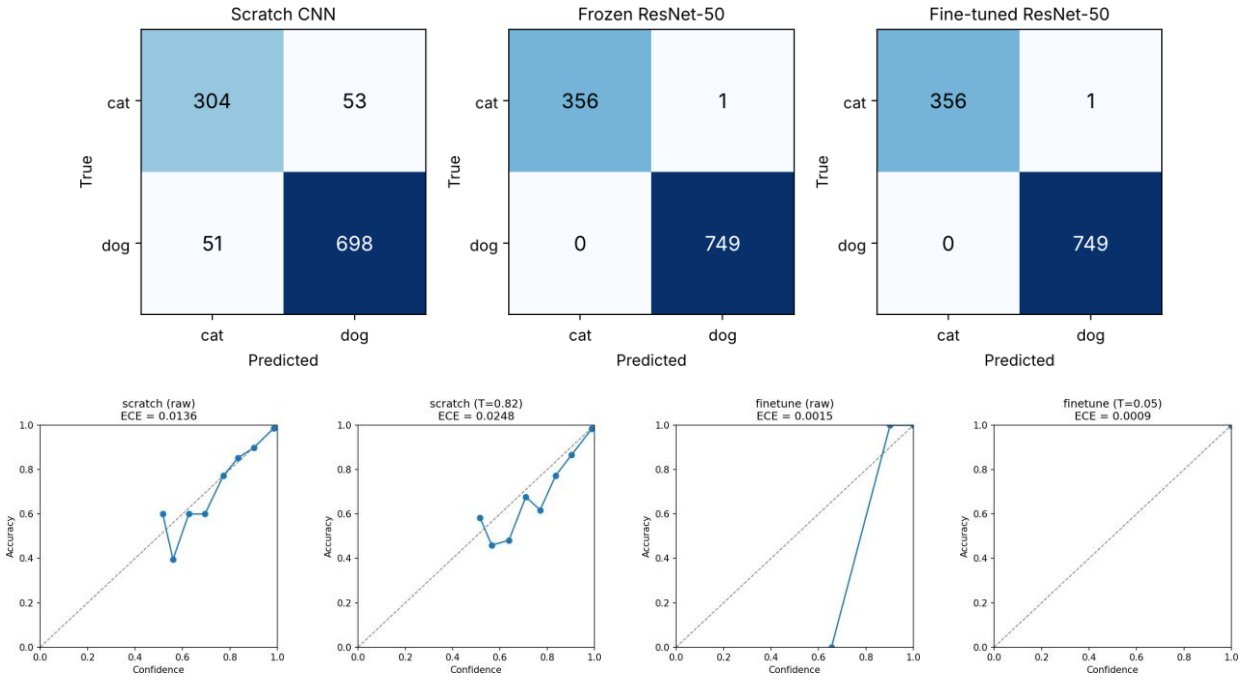


Figure 4: (Top) confusion matrices of the three models (rows true, columns predicted). (Bottom) reliability diagrams, left to right: scratch raw, scratch with $T = 0.82$, fine-tuned raw, fine-tuned with $T = 0.05$. Test split; input 224 × 224 px; argmax decision, no threshold; 15 confidence bins.

Most confused pair: cat → dog. The selected model makes one error in 1,106 images, a black Bombay cat predicted as a dog; the frozen model's single error is a Sphynx cat. The scratch CNN confuses both directions about equally (53 cats predicted as dogs, 51 dogs predicted as cats).

Over- or under-confidence: the raw scratch CNN is close to the diagonal above 0.75 confidence and over-confident in the 0.55–0.70 bins. Temperature scaling fitted $T = 0.82 < 1$ on the validation split, which sharpens the probabilities and made the test calibration worse (ECE 0.014 → 0.025, over-confident across 0.55–0.85). For the fine-tuned model the validation split is almost perfectly

separated, so the fitted temperature runs to the lower clip ($T = 0.05$ is a bound, not a fitted value) and the model is confident close to 1.0 on nearly every image. With one error in 1,106 images the change in ECE (0.0015 → 0.0009) cannot be told apart from sampling variation. Calibration on a near-separable validation split is therefore ill-posed. The calibrated confidence is not treated as a reliable probability, and the referral threshold (0.90) is provisional, not a deployment-ready operating point: it will be fixed and measured on a harder, independent calibration set, with the final test set kept closed (plan below).

3.5 Error analysis

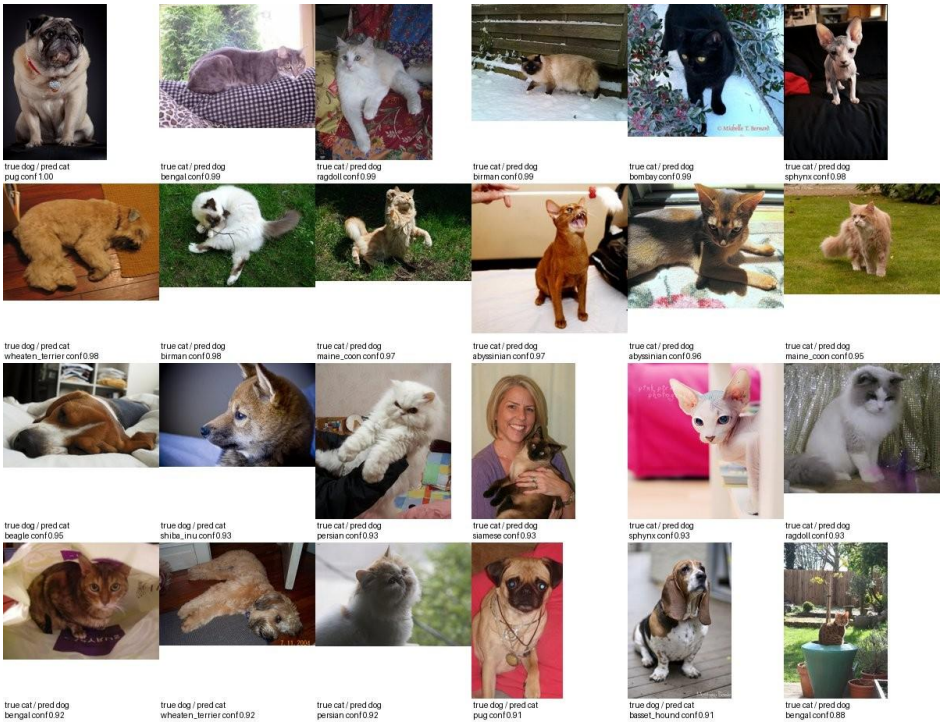


Figure 5: Gallery of the most confident misclassified images of the scratch CNN (true label, predicted label, breed, confidence). Test split; input 224 × 224 px; argmax decision, no threshold.

Error pattern	Examples	Likely cause (data, label, model)
Long-haired and colour-point cats predicted as dogs (scratch)	Ragdoll 5, Birman 5, Persian 4, Maine Coon 4 of 30 each	Model: fur texture and body shape resemble dogs; the small CNN has not learnt finer cues (still under-fitted)
Hairless and black cats predicted as dogs	Scratch: Bombay 7 of 28, Sphynx 6 of 30; fine-tuned: 1 Bombay; frozen: 1 Sphynx	Data: unusual coat appearance and low contrast; few such cats in training
Dogs lying down or curled up predicted as cats (scratch)	Pug 5, wheaten terrier 5, beagle 3, basset hound 3 of 30 each	Data and model: pose hides the body shape; few curled-up dogs
Label noise	0 species errors in the 200-image audit	Label: unlikely as a main cause, but not ruled out for individual images

Counts are errors per breed on the test split. Of the 24 most confident scratch errors, 17 are cats predicted as dogs. The pose and fur explanations come from inspecting the gallery and are hypotheses, not measured effects.

3.6 Data-improvement plan

Finding from the error analysis	Planned action	Expected effect
Hairless, black and colour-point cats dominate the cat-to-dog errors; dogs outnumber cats 2:1	Collect and label more cats of these types and rebalance the species	Fewer cat-to-dog errors and higher cat recall. Data and model choices are made on the training and validation splits only; the effect is measured once on a new, independent test set, because the current test split has already been read for the final models
Curled-up dogs, puppies and small or partly visible animals are rare in Oxford-IIIT Pet; on the team photos, 9 of the 12 errors are dogs predicted as cats, several of them puppies	Collect new harder photos (puppies and kittens, lying or curled-up animals, clothes, several animals, small animals); the 401 team photos stay an external test set and are not added to training	Better generalisation to shelter-style photos, checked on the untouched external test set
Temperature runs to its lower bound on a near-separable validation split	Fit and compare calibration strategies (including a non-sharpening $T \geq 1$ constraint) on a new, harder calibration set kept apart from the external test set	A confidence score whose referral threshold is meaningful
Four loose derived body boxes in the 300-box review	Re-derive these boxes and add a box-quality check to the pipeline	Cleaner box labels for the Phase 2 detector

4 Reproducibility (8 points)

Item	Your answer
Repository and commit hash	Folder `smart-pet-vision` in the submission archive (snapshot of 28 October 2026; SHA-256 of every file in `hANIFEST_SHA256. txt`, which serves as the fixed identifier of the submitted version). The project was developed in a shared Google Drive folder, not git, so there is no commit hash; the reported results are traced through the W&B runs `gh7ayx6h`, `xsz5afr9`, `8nvxbsn2` (group `full-rerun-2026-10-18`) and the executed notebooks in `notebooks/executed`.
Configuration file used for the reported runs	`configs/config. yaml` (single file for all notebooks)
Best checkpoints	`runs/classification/{scratch, frozen, finetune). pt` (epoch of best validation macro-F1), SHA-256 in the manifest. Reloading them reproduces the submitted test accuracy of all three models exactly (0 images differ; notebook `02b`, `runs/supplement/checkpoints. j son`)
Random seeds	42 for Python, NumPy, PyTorch (CPU and CUDA), `PYTHONHASHSEED`, the split and the data loader
Requirements file (pinned versions)	`requirements. txt`: all 19 dependencies pinned to exact versions of the 18–19 October run (e.g. torch 2.11.0, torchvision 0.26.0, wandb 0.28.1)
Command(s) to reproduce the main result	Extract the submission archive, which keeps `SmartPetVision_TeamData` next to `smart-pet-vision` (steps in `READhE_SUBhISSION. md`; Phase 1 does not read the team data). In Colab with a T4: set `SPV_REPO` to `smart-pet-vision`, then run `notebooks/01_data_preparation. ipynb` and `notebooks/02_classification. ipynb`; compare `runs/classification/comparison. csv` with the expected values in `REPRODUCE. md`. W&B login uses `vandb login` or a Colab secret; the key is never written in a cell or a file.
Where the data is obtained (not committed)	Oxford-IIIT Pet is downloaded by the loader from the official VGG page to the Colab runtime disk (`/content/data`), never to the repository. The team photos are in the folder `SmartPetVision_TeamData` next to the repository in the submission archive, read through `paths. team`: `. . /SmartPetVision_TeamData` in the configuration; they are shared only with the team and the instructor
Reproduced by (team member) and result	Yasser Alhajlah, 23 October 2026, clean Colab T4 runtime on a copy of the folder (split rebuilt from scratch): every dataset and classification value identical to the submitted run (W&B group `reproduction-2026-10-23`). The team data were moved out of the repository afterwards (26 October); no model, split or metric changed

5 Submission checklist

- ☒ Corrupt files and duplicates handled, with counts reported
- ☒ Split is grouped by the source of correlation; the grouping key is justified
- ☒ Label audit reported with a confidence bound
- ☒ Dataset card states licences, limitations and privacy measures
- ☒ Scratch and fine-tuned classifiers compared in W&B
- ☒ Per-class metrics, confusion matrix and calibration (ECE) reported
- ☒ Test set used only for final evaluation, not model selection
- ☒ Data-improvement plan follows from the error analysis

- ☒ Fixed seeds, single configuration, pinned requirements; no hard-coded paths
- ☒ Every figure states split, input resolution and thresholds
- ☒ No data or personal information committed to the repository (team photos in a separate folder outside the repository; Oxford-IIIT Pet downloaded at run time)

6 Self-assessment

Criterion	Points	Self-score	Strongest evidence (section)	Known gap
Data foundations and dataset card (M1)	12	12	Group-aware stratified split with zero near-duplicate (MD5/pHash) crossings; identity leakage measured (effect ≤ 0.001 on the selected model); label and box audits with bounds; privacy measures and full audit of the team photos (2.2–2.6)	No identity labels, so identity leakage is bounded by a similarity proxy, not excluded
Classification (M2–M3)	20	20	Three runs in W&B; full learning curves (training and validation loss and accuracy); best checkpoints; per-class metrics, confusion matrices, ECE and reliability diagrams; McNemar test; error table and plan (3.1–3.6)	Calibration ill-posed on a near-separable validation split, so the confidence threshold is provisional; validation used for three decisions
Reproducibility	8	8	Single configuration, seed 42, pinned requirements, data kept outside the repository; full rerun by a second member matched every value; checkpoints reproduce the test scores; logging rerun matched every logged value (4, 3.2)	Developed in a shared Drive folder, not git, so no commit hash; results traced through W&B run IDs and SHA-256 checksums
Total	**40**	**40**		

7 Obstacles and how they were addressed

Obstacle	Module	Impact on the work	Action taken	Status
Oxford-IIIT Pet has head boxes only, no full-body boxes	M1	No box labels for the Phase 2 detector and cut-out	Body boxes derived from the trimaps; 300 random boxes reviewed (4 loose, 1.3%)	Resolved
14 images with an empty trimap	M1	No usable mask or body box for them	Dropped during cleaning and reported	Resolved
No identifier for the individual animal	M1	The same pet may appear in more than one split	Near-duplicate groups kept in one split; identity leakage measured with nearest same-breed training neighbours: the selected model's accuracy stays 0.997–0.999 on test images without a close match (2.3)	Resolved (bounded)
Near-separable validation split	M2–M3	Temperature scaling runs to its bound; ECE uninformative	Limits reported; calibration on a harder, separate set planned	Open
Privacy and provenance of team photos	M1	GPS and other metadata in the files; some photos looked like stock or generated images	Owner consent; metadata removed; 13 photos excluded	Resolved
Team photos stored inside the project folder (`data/team`)	Repro.	Data sat in the code repository	Moved to the folder `SmartPetVision_TeamData` next to the repository; configuration and `REPRODUCE.md` updated	Resolved
Rerunning a notebook overwrites `runs/` on the shared Drive	Repro.	A reproduction could overwrite the submitted results	Reproduction run on a copy via `SPV_REPO`; procedure in `REPRODUCE.md`	Resolved

A Dataset card

This appendix reproduces `docs/dataset_card.md` from the `smart-pet-vision` folder. Figures come from `runs/data/dataset_card_values.json` (notebook 01), `runs/data/reviev/` and the files of the team-data folder `SmartPetVision_TeamData`, which sits next to the repository folder, not inside it. Last updated from the full rerun of 18–19 October 2026.

Sources

Set	Content	Use	Licence
Oxford-IIIT Pet (Parkhi et al., 2012)	37 breeds (12 cat, 25 dog); species and breed labels, head box, pixel trimap	train / val / test	CC BY-SA 4.0
Team-collected	401 photos of cats and dogs (178 cat, 221 dog, 2 mixed); 503 body boxes; polygon masks	external test only	supplied by team members (104 taken by pet owners), annotated by the team, owner consent; not redistributed

Cleaning

- Unreadable files, images under 32 px and images without a usable trimap are dropped: 14 dropped (empty trimap).
- Byte-identical files (MD5) keep one copy: 0 exact duplicates found.
- Near-duplicates (pHash Hamming distance ≤ 4): 45 pairs, merged transitively (union-find) into 41 groups covering 84 images (largest group 3).
- After cleaning: 7,335 images, 2,366 cats, 4,969 dogs, 37 breeds. Median image size 500 × 375.

Split

70/15/15 by image count, stratified by breed, whole duplicate groups assigned to one split. Counts: train 5,123 / val 1,106 / test 1,106. Leakage audit: groups crossing splits = 0, MD5 crossing splits = 0 (passed).

Split	Images	Cats	Dogs	Cat share	Smallest breed
Train	5,123	1,652	3,471	32.2%	128
Val	1,106	357	749	32.3%	28
Test	1,106	357	749	32.3%	28

Residual risk: the dataset has no per-animal identifier, so different photos of the same pet can fall in different splits. Its effect was measured with the nearest same-breed training image of every test image (ImageNet ResNet-50 features): removing the test images with a close training match (similarity ≥ 0.95, 22 images, down to ≥ 0.75, 790 images) leaves the fine-tuned accuracy at 0.997–0.999 (`runs/supplement/identity_effect.csv`). Similarity is a proxy, so identity leakage is bounded, not excluded. The team-collected set (photos supplied by team members or pet owners and reported to have no MD5/pHash match with Oxford-IIIT Pet) is an external test of generalisation to different photo sources. It cannot establish identity-disjointness within Oxford-IIIT Pet or prove zero possible pretraining overlap. Test accuracy is reported with a 95% Wilson interval (fine-tuned ResNet-50: 0.999, CI 0.995–1.000).

Derived annotations and audits

- **Body mask** = trimap foreground + border. The border band is ambiguous; segmentation is also scored with it ignored.

- **Body box** = tight box around the body mask. Review of 300 random images: 4 flagged (1.3%, 95% CI 0.5–3.4%), all loose boxes around a small animal; none misses the animal (`runs/data/reviev/boxes/reviev.csv`).
- **Label audit** of 200 random images: 0 species errors (0%, 95% upper bound 1.5%); 4 breed labels doubtful (2.0%, 95% CI 0.8–5.0%) (`runs/data/reviev/labels/reviev.csv`).
- Both audits were done visually on the contact sheets with AI assistance on 19 October 2026 (`reviever` column) and confirmed by the team (`ovner_confirmed` column). Notebook 01 prints the summary from these files. Upper bound: rule of three; intervals: Wilson.

Known limitations

- Mostly one large, centred animal per image (1,105 of 1,106 test boxes are 'large'); web and pet-site photos are cleaner than shelter photos, so small or partly visible animals are under-represented.
- Class imbalance: about two dogs per cat; handled by breed stratification and macro-F1, no class weights.
- The same individual animal may appear in more than one split (no identity labels); measured effect on the selected model at most 0.001 (similarity proxy).
- Possible overlap between ImageNet pretraining images and Oxford-IIIT Pet (both web-collected) cannot be excluded.
- Breed is not a protected attribute, but per-breed error rates are reported because some breeds look alike across species.

Team-collected set

- **Collection.** 414 photos received from team members; 13 excluded as suspected stock or generated images; 401 kept. Who took the kept photos: 104 were taken by the pets' owners and sent as screenshots (unedited, per team statement); the other 297 were supplied by team members (170 with phone-camera EXIF, 127 without camera metadata). `source_map.csv` does not record which of these 297 a team member photographed personally and which a team member received from an owner, so the set is described as team-collected and team-annotated, not as photographed entirely by the team. Owner consent reported by the team for all 401; signed or recorded permission and provenance evidence is retained privately and not added to the code repository.
- **Processing (same M1 protocol).** EXIF orientation applied, then all metadata including GPS removed; resized to a maximum side of 2048 px; MD5 and pHash checked within the set and against Oxford-IIIT Pet (no exact or near duplicates found); quality audit for blur, size and black bars (`quality_audit.csv`). No identifiable faces; animals not disturbed.
- **Annotation.** CVAT (CVAT for images 1.1). An automatic draft (detector boxes and segmentation polygons) was reviewed and corrected in CVAT by the four team members, one part each (part 1: 101 images; parts 2–4: 100 images each). Images changed from the draft: part 1: 4, part 2: 9, part 3: 100 (46 with redrawn polygons), part 4: 5. Four species labels were corrected by the annotators (team_0144, team_0354, team_0391, team_0392) and four more after a visual review of the classifier–label disagreements (team_0065, team_0076, team_0097, team_0209; `label_corrections.csv`). Per-part notes are in `annotation/done/`.
- **Full audit (notebook 02b,** `runs/supplement/team_audit.json`**).** All 401 photos readable, each with a mask of the same size and a label (401/401/401); 503 boxes, matching `n_animals` for every photo; no EXIF metadata left; no photo above 2048 px; no exact duplicates; median mask share 0.28; species 178 cat, 221 dog, 2 mixed. Every photo is rendered with its boxes in `audit_02b/team_sheet_00-08.jpg` inside the team-data folder (outside the repository).
- **Labels used in evaluation.** `labels.csv` (image, species, n_animals), `team_boxes.csv` and `masks/` (union of the polygons per image). The 2 mixed cat-and-dog images are excluded from classification metrics.
- **Use.** External test only: never used for training, threshold selection or model selection.
- **External test result (classifier).** Fine-tuned ResNet-50 applied to the detector crop in the integrated pipeline (notebook 05, Phase 2 results; `runs/deploy/team_metrics.json` in the Phase 2 submission): accuracy 0.970 (387 of 399 single-species photos; Wilson 95% CI 0.948–0.983), macro-F1 0.970; 0.960 before the four post-review label corrections, which came from reviewing only classifier–label disagreements and can bias accuracy upward.

- **Storage.** The processed set (images, masks, annotations, label and audit files) is kept outside the code repository in `SmartPetVision_TeamData`, shared only within the team and, for grading, with the instructor; the code reads it through `paths. team: .. /SmartPetVision_TeamData` in `configs/config. yaml`. Unprocessed originals, which may carry camera metadata, are kept privately and are never shared. No team photo file is stored in the repository or redistributed.